Regional Failure Playbook — me-central2 (Dammam)

Region: me-central2 (Dammam, Saudi Arabia) — mandatory per SDAIA + NELC. **Scope:** What happens if the entire GCP me-central2 region degrades or goes offline, and how to recover. **Verified:** 2026-08-17

1. What lives in me-central2

Every production resource is in me-central2 (no cross-region redundancy):

Resource	Type	Config	Recovery source if lost
`physioguidance-db-me`	Cloud SQL (Postgres 15)	Zonal (me-central2-a), PITR ON, 7-day automated backups, deletion-protected	Automated backups (Cloud SQL) + PITR
`physioguidance-redis`	Memorystore Redis 7	BASIC 1 GiB, no persistence	Nothing — cache-only; app rebuilds it (see redis-persistence-risk.md)
`physioguidance-api`	Cloud Run service	Regional	Source (git) — redeploy
`physioguidance-api-worker`	Cloud Run service	Regional	Source (git) — redeploy
`physioguidance-web`	Cloud Run service	Regional	Source (git) — redeploy
`physioguidance-cms` (Strapi)	Cloud Run service	Regional	Source (git) — redeploy + DB restore
`physioguidance-malware-scanner`	Cloud Run service	Regional	Source (git) — redeploy
`physioguidance-public`	GCS bucket (public assets)	Regional me-central2	See §5
`physioguidance-private`	GCS bucket (user uploads)	Regional me-central2	See §5
`physioguidance-trainers`	GCS bucket (trainer docs)	Regional me-central2	See §5
`physioguidance-cms` bucket	GCS bucket (Strapi media)	Regional me-central2	See §5
`physioguidance-quarantine`	GCS bucket (malware isolation)	Regional me-central2	Discardable
`physioguidance-backups`	GCS bucket (app backups)	Regional me-central2	⚠ Co-located with data — see §5.1
25 Cloud Run Jobs	Batch workloads	Regional	Source (git) — redeploy
Cloud Tasks queues	Event scheduling	Regional	State auto-rebuilds
Cloud Scheduler jobs	Time-triggered	Regional	Source (git) — redeploy
Eventarc triggers	Bucket → scanner	Regional	Source (git) — redeploy

Resource	Type	Config	Recovery source if lost
Secret Manager secrets	Credentials	Auto-replicated within region	Manual re-create if bucket-region wiped

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2. Failure modes ordered by likelihood

2.1 Single-zone incident within me-central2 (most likely)

- **What happens:** One AZ (typically me-central2-a) becomes unreachable.
- **Impact today: Cloud SQL goes offline** (it's ZONAL in me-central2-a). Cloud Run services are multi-zone and stay up but return 5xx because the DB is down.
- **Duration expected:** Minutes to hours (per GCP SLA).
- **Recovery:** Wait for GCP or restore DB from backup into a different zone. See §4.
- **RTO (Recovery Time Objective) today:** ~2 hours (DB restore from backup).
- **RPO (Recovery Point Objective) today:** ~5 minutes worst case (PITR granularity), typically <1 minute.

2.2 Full region outage (rare but possible)

- **What happens:** All of me-central2 (Dammam) is unreachable.
- **Impact:** Total platform offline. Cloud Run, Cloud SQL, Redis, GCS, Secret Manager — all unavailable.
- **Duration expected:** Hours (historical GCP regional incidents rarely exceed 4 hours).
- **Recovery:** WAIT for GCP recovery — we cannot cross-region failover, and PDPL/SDAIA forbid moving customer data outside Saudi Arabia.
- **RTO today:** = GCP recovery time.
- **RPO today:** = last automated backup (up to 24 hours old) — but we can't restore into a different region either (PDPL forbids), so recovery = wait for region.

2.3 Data corruption / accidental deletion (not a regional event but same playbook)

- Handled by PITR — see backup-system-decision.md and §4.2.

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3. What's protecting us today

- **Cloud SQL automated backups** — daily snapshots, 7-day retention, encrypted at rest.

- **PITR (Point-in-Time Recovery)** — enabled; can restore to any timestamp in the last 7 days at ~5-minute granularity.
- **Deletion protection** on Cloud SQL — a rogue `gcloud sql instances delete` call fails.
- **App backups bucket** — `physioguidance-backups` stores logical DB dumps + Strapi content dumps (see backup-system-decision.md).
- **Source-controlled infrastructure** — every Cloud Run service, job, scheduler, IAM binding is defined in `infra/` scripts. Full rebuild-from-git is possible.
- **Terraform / gcloud scripts in git** — no hand-clicked infrastructure that would be lost with an operator.

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4. Recovery procedures

4.1 Single-zone Cloud SQL outage — restore to a different zone

When GCP marks me-central2-a as impaired but me-central2-b is healthy:

```
# 1. Confirm current instance is unreachable (not just slow)
gcloud sql instances describe physioguidance-db-me --format="value(state)"

# 2. Create clone in me-central2-b from latest automatic backup
gcloud sql instances clone physioguidance-db-me physioguidance-db-me-recovery \
  --preferred-zone=me-central2-b \
  --project=physioguidance

# 3. Wait for RUNNABLE (typically 15-30 min for a 10GB instance)
gcloud sql instances describe physioguidance-db-me-recovery \
  --format="value(state)" -w

# 4. Rotate DB connection secret to the new instance connection name
# (Secret Manager: DATABASE_URL) then redeploy backend + worker + CMS
# via infra/backend/deploy.sh, infra/worker/deploy.sh, infra/strapi/deploy.sh

# 5. When me-central2-a is healthy again, either:
#    (a) keep running on -recovery and rename, OR
#    (b) clone -recovery back to a fresh instance in me-central2-a
```

Estimated RTO: 45-90 minutes end-to-end (clone + secret rotation + 3 redeploys).

4.2 Data corruption or accidental delete — PITR restore

```
# Restore to a NEW instance at the timestamp just before the incident.
# NEVER PITR-restore into the running production instance in place.
gcloud sql instances clone physioguidance-db-me physioguidance-db-restore-<date> \
  --point-in-time="2026-08-17T13:45:00Z" \
  --project=physioguidance

# Then either promote it (rotate secret + redeploy) or export the specific
# rows you needed and re-import into production.
```

See backup-system-decision.md for the "restore-to-clone only, never in-place" policy.

4.3 Cloud Run service loss

```
# All Cloud Run services + jobs are code — just redeploy from a healthy machine:
./infra/backend/deploy.sh
./infra/worker/deploy.sh
./infra/frontend/deploy.sh
./infra/strapi/deploy.sh
./infra/malware-scanner/deploy.sh
```

Cloud Run deploys are idempotent — running them against a partially-recovered region re-creates any missing service.

4.4 Full-region outage

1. **Wait.** Check https://status.cloud.google.com and https://status.cloud.google.com/regional/asia — GCP posts region incidents there.
2. **Communicate.** Post a maintenance banner via the pre-existing `/dashboard/admin/settings` → Scheduled Maintenance module (this itself needs backend up, so alternately post via Sentry status page or a static holding page hosted on the domain registrar).
3. **Do NOT attempt cross-region failover.** PDPL/SDAIA prohibits customer PII outside Saudi. `us-central1` etc. are not legal recovery targets.
4. **Preserve local backups.** If the outage extends >24h, verify `physioguidance-backups` bucket contents are still accessible via `gcloud storage ls` from a healthy region — the bucket is regional in me-central2, so bucket access may also be impaired.
5. **When GCP restores the region**, run the checklist in §6.

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5. GCS bucket resilience

All buckets are **regional in me-central2**. GCP's 11-9s durability applies within a region — the bucket itself does not lose objects. But bucket **availability** depends on region health: a full-region outage means the bucket is unreadable, not gone.

Objects come back when the region comes back. No object-level recovery needed.

5.1 Known risk: `physioguidance-backups` is co-located with the data it backs up

The backups bucket lives in me-central2 alongside Cloud SQL. If the region is offline, backups are unreadable during the outage.

Why we accept this:

- PDPL/SDAIA — customer data must stay in Saudi. There is no second Saudi region.
- The purpose of these backups is data-corruption recovery, not region-outage recovery.
- Region-outage recovery relies on **Cloud SQL's own backups** (managed by GCP, replicated within the region across zones per GCP internals we don't control).

When to revisit:

- If Google launches a second Saudi region: enable dual-region for the backups bucket.
- If a compliance auditor requires geographically-separated backups: negotiate with SDAIA for encrypted cross-region storage (out of scope today).

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6. Post-outage verification checklist

After GCP marks the region recovered, run in order:

- ☐ `gcloud sql instances describe physioguidance-db-me --format="value(state)"` → `RUNNABLE`
- ☐ `gcloud redis instances describe physioguidance-redis --region=me-central2 --format="value(state)"` → `READY`
- ☐ All 5 Cloud Run services return HTTP 200 on `/health/ready` (or `/` for web)
- ☐ `curl https://api.physioguidance.org/health/ready` → `{"db":"ok","redis":"ok"}`
- ☐ Sentry error rate returns to baseline within 15 min (spike expected during recovery is normal)
- ☐ `/dashboard/admin/monitoring` shows all services green
- ☐ Test end-to-end flow: log in as test student → view a course → check certificate download works
- ☐ Test payment flow: create a test order via one payment gateway (use sandbox mode)
- ☐ Confirm no stuck webhooks in `/dashboard/admin/finance` older than 15 min

- [ ] Confirm Cloud Tasks queue depth is draining (Cloud Console → Cloud Tasks)
- [ ] Post `all-clear` update on maintenance banner

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7. What's deliberately NOT done today (and why)

7.1 Multi-region hot standby

- **Blocked by:** PDPL — no second Saudi region exists.
- **Would provide:** Zero-downtime regional failover.
- **Action:** N/A until a second me-* region launches.

7.2 Cross-region backup replication

- **Blocked by:** PDPL — data must stay in Saudi.
- **Would provide:** Backup access during a full-region incident.
- **Alternative today:** Rely on Cloud SQL's internal cross-zone backup replication (managed by GCP) + accept the region-outage window.

7.3 Regional HA Cloud SQL (`REGIONAL` availability type)

- **Cost delta:** ~$45/mo (upgrade from `db-g1-small` ZONAL to `db-g1-small` REGIONAL with a standby in me-central2-b).
- **Would provide:** Automatic failover if me-central2-a goes down (RTO drops from ~90 min to ~2 min for a single-zone incident).
- **Recommended: Yes, when budget permits.** This is a legitimate 15x RTO improvement for single-zone incidents (the most common failure mode) at a modest cost. Not blocking today — but should be first upgrade after the platform is generating revenue.

7.4 Runbook rehearsal

- Recovery procedures in §4 are **not rehearsed**. First real execution will take longer than the estimates given.
- **Recommended:** Schedule a quarterly DR test — clone DB to a `-drill` instance, run a dummy failover, tear down. First drill: after handover complete.

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8. Summary for Amna

- **Single-zone outage (Cloud SQL fails in me-central2-a):** ~90 min recovery via PITR clone into me-central2-b. Documented in §4.1.
- **Full-region outage:** wait for GCP. Cannot cross-region due to PDPL. Historical GCP regional outages rarely exceed 4 hours.
- **Data corruption or accidental delete:** PITR restore to a fresh instance in <30 min. Documented in §4.2.
- **Redis loss:** cache repopulates automatically; queues have reconciliation. See redis-persistence-risk.md.
- **RTO today:** ~90 min for zone incident; = GCP-region-recovery-time for region incident.
- **RPO today:** <5 min for PITR; up to 24h for full-region incident (limited by daily backup cadence).
- **Highest-value upgrade:** REGIONAL Cloud SQL availability type (~$45/mo, 15x RTO improvement for the most common failure mode). Recommended after handover.
- **No action required today.**